

02 Measure a room — the AR path

Twice you reported this doing nothing. This is not a test of how accurate it is. It is a test of whether it gets from a live camera picture to a closed room **at all**. Take a tape anyway — step 9 is worth having.

BEFORE YOU START

A room with **four square corners** and the floor visible all the way round — a bedroom or your garage. Not an L-shape, not the kitchen with an island. Lights on. You need to be able to walk the whole perimeter without climbing over anything.

WHAT TO DO

- 1 First screen → **Measure a room**. The camera comes up.
- 2 Watch the sentence at the top. It will say `Starting up...` then most likely `Move the phone slowly from side to side to start tracking`. Do that — slowly, a couple of feet each way.
- 3 When tracking settles, it says `Lay the phone flat on the floor and tap Set floor`. **Put the phone face up on the floor**, flat, and tap **Set floor**.
If your phone found the floor by itself it says `Floor found` instead and you can skip this — both are a pass.
- 4 Pick the phone up. It should now read `Floor set – phone is 3'4" above it` (your height, whatever it is) and the instruction changes to `Point at the foot of a corner and tap`. **The dot in the middle of the screen goes solid**.
- 5 Point the dot at **the foot of a corner** — where the two walls meet the floor, not up at eye level — and tap the big round button.
- 6 Walk to the next corner. It says `Walk to the next corner and tap`. Tap it. Then `Keep going round. Tap the first corner again when you get back to it`.
- 7 Wall lengths appear along the bottom as you go — `wall 1`, `wall 2` ... **Read them as they land**. That is the app measuring in front of you.
- 8 Back at the first corner it says `You are back at the first corner – tap it again to finish`. Tap it. **Done** is now tappable (it needs four corners). Type the room name into `What is this room?` and tap **Done**.
- 9 **Now the tape**. Measure two walls that meet at a corner — one running each way — and write them here against what the app said:

WALL	WHAT THE APP SAID	YOUR TAPE	OFF BY
wall 1			
wall 2			

Corner-tapping by hand is not LiDAR. A couple of inches out is normal and is the whole reason the app refuses to call it measured. What matters is that it *finished*.

PASS

The dot went solid, four taps landed, lengths appeared, **Done** worked, and the room opened on the review screen.

FAIL – AND WHAT TO SEND ME

The exact sentence on the screen, word for word. It is different for every reason this can fail, and that sentence is the diagnosis. Do not paraphrase it.

IF IT STALLS, THIS IS WHAT IT IS TELLING YOU

ON SCREEN	WHAT IT MEANS	WHAT TO DO
Slow down – the phone is moving too fast to track	You are sweeping it.	Move at walking pace or slower.
Not enough detail to track	Blank walls, bare floor, low light.	More light. Point at something with pattern in it.
Finding where it is again – hold still a moment	It lost itself, usually a fast turn.	Stand still. It comes back.
It is not flat enough to tell where the floor is.	Phone was tilted on Set floor .	Flat on the floor, screen up, then tap.
Point further down – the middle of the screen has to be on the floor	You aimed at the corner at eye level.	Aim at the foot of the corner.
This device cannot track its position well enough	Not a real phone, or too old.	A simulator will never pass this. Use the iPhone.

Test 2 exists because the first two fixes were real and were not the whole thing. The floor height is now read straight off the current camera frame, which nothing else can take away from it — and if the phone never finds the floor on its own, you tell it where the floor is by putting the phone on it.

20 Changing the room

Everything this app promises comes down to one thing: **a number you moved by hand must never be able to pass as a number you measured**. Step 4 is the test. If it ever fails, nothing else on any of these sheets is worth anything.

START FROM THE ROOM YOU JUST WALKED, OR ANY SAVED SCAN

- 1

Rename the room. Top of the plan → **Rename**. Give it the name you would say out loud — “Gilbert’s kitchen”.
Check: that name now heads the drawing, the takeoff and the claim document. It used to say `garage.json`.

2

Name a wall. Tap a wall on the plan → **Change this wall** → `Call it something`. Type “behind the washer”.
Check: anything marked on that wall moved with it.

3

Move one wall. Same panel → `Move it – without a tape`. Type a length a foot or two off what it says — then **Move it**.
Check three things: the wall goes exactly where you put it; the rest of the room gives way around it; and the wall turns ■ **violet** with `moved by hand` beside the number. The legend under the plan grows a `Moved by hand` entry.

4

Now move all four walls. One at a time, same way.
- THIS IS THE TEST

With every wall in the room moved by hand, the plan must **still** say
SCANNED – no wall here has had a tape on it. These numbers will move. and the app must **still** ask you to put a tape on one wall running each way.
- FAIL

If anywhere — plan, takeoff, price, client file, claim — now says the room is measured, **stop and tell me immediately**. Send the scan. That is the one bug that would make this app dishonest.
- 5

Put a real tape on that wall. Measure it properly, then `It has had a tape on it` → type the reading.
Check: the wall goes ■ **black**, says measured, and the whole room re-solves around it.

6

Try to drag that same wall again.
Check: it will not let you. The panel now offers only **Take the tape reading off** — because dragging a measured wall would quietly replace a measurement with a guess while the plan still said somebody measured this room.

THEN THE SHAPE EDITS – EACH HAS ONE THING TO WATCH

WHAT YOU TAP	WHAT IT IS FOR	WHAT MUST HAPPEN
Cut it in two	A pony wall meeting a full-height one.	It asks how high the second piece is, and refuses if your answer makes the two pieces identical. That refusal is correct — two walls in line alike in every way are one wall written twice.
Notch the corner after it	An alcove the scanner flattened.	It puts two walls in, not one — a room with square corners always has an even number of sides. The room does not change size: the two walls beside the corner give up the length.
There is no wall here at all	A side the scan invented.	The line under the plan names which walls moved and by how far . Undo is one tap.

WRITE DOWN ANYTHING THAT SURPRISED YOU

STEP	WHAT YOU EXPECTED	WHAT HAPPENED

COLOURS ON THE PLAN

■ Measured

■ Scanned

■ Moved by hand

■ No wall here

■ Something in the way

A dragged wall re-solves the room exactly as a tape reading does — same solver, same arithmetic. The only difference is what the room is then allowed to claim about itself, and that difference is the product.